

	$V = E - Ir$ $5.4 = 9.0 - 1.8(r)$ $r = 2.0 \, \Omega$
(b)(iii)	Add a switch in correct location.
	When the switch is used, it will result in an open circuit between the resistors and the cell. Since no current flows through the cell, the terminal potential difference is the same as the emf.
(c)	$\text{p.d. across } 2R = \left(\frac{25 + 25}{3(25) + 15} \right) \times E$
	Since galvanometer reads zero, p.d. across $2R = \text{p.d. across } R_c$ $\left(\frac{25 + 25}{3(25) + 15} \right) \times E = \left(\frac{R_c}{R_c + 15} \right) \times E$
	$R_c = 18.8 \, \Omega$

5(a)	
(b)(i)	PSYV
(b)(ii)	Electrons experience a magnetic force.
	The accumulation of the electrons causes a potential difference to be set up.
(iii)	$F_e = F_B$
	$e \frac{V_H}{d} = evB$ $V_H = 0.020 \times 56 \times 0.28$ $= 0.31 \text{ V}$
(b)(iv)	Deflection would be opposite, so positive charges will accumulate on opposite face as before.
	Hence, polarity of the hall voltage will not change.

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6(a)	Apply Charles' law
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